

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2023-24

Paper: CS-102 — Object Oriented Programming

Time: Three hours

Maximum Marks: 70

Instructions: Attempt five questions including Question No. 1, which is compulsory. All undefined symbols have their usual meanings.

Question 1. Attempt all questions: [2 × 9 = 18]

- (a) Differentiate between procedural programming and object-oriented programming.
- (b) What is dynamic binding in C++?
- (c) Explain the purpose of a destructor.
- (d) What is a reference variable? Show its usage with an example.
- (e) What is a copy constructor? Write its prototype.
- (f) Can a virtual function be declared inline? Why or why not?
- (g) Find the output of the following code segment:

```
#include <iostream>
using namespace std;
class Test {
    static int count;
public:
    Test() { count++; }
    static int getCount() { return count; }
};
int Test::count = 0;
int main() {
    Test t1, t2, t3;
    cout << "Count=" << Test::getCount() << endl;
    return 0;
}
```

- (h) Identify and explain the compile-time error in the following code block:

```
#include <iostream>
using namespace std;
class Box {
    double width;
public:
    Box(double w) : width(w) {}
};
void printWidth(Box b) {
    cout << b.width << endl;
}
```

- (i) What is the use of `this` pointer in C++?

Question 2.

- (a) What is encapsulation? How is it implemented in C++? Compare private, protected, and public access specifiers. [6]
- (b) Explain the concept of inline functions. What are the rules and situations where inline function expansion is ignored by the compiler? [4]
- (c) Discuss function overloading with a suitable code example. [4]

Question 3.

- (a) Explain the dynamic memory allocation operators `new` and `delete` in C++ with syntax and examples. Differentiate between `delete` and `delete[]`. [6]
- (b) What are friend functions and friend classes? Discuss their advantages and disadvantages with a suitable example. [8]

Question 4.

- (a) What is constructor overloading? Write a C++ class representing a `Complex` number with default, parameterized, and copy constructors. [8]
- (b) Write the rules for overloading operators in C++. How does overloading a binary operator using a friend function differ from overloading it using a member function? [6]

Question 5.

- (a) Write a C++ program to overload the binary addition operator `+` to add two `Matrix` objects. [8]
- (b) Explain unary operator overloading with an example of overloading pre-increment `++` and post-increment `++` operators. [6]

Question 6.

- (a) What are the different types of inheritance supported in C++? Explain the concept of virtual base class in multiple inheritance with a diagram and code snippet to show how ambiguity is resolved. [8]
- (b) Discuss the difference between compile-time polymorphism and runtime polymorphism. How is runtime polymorphism achieved in C++? [6]

Question 7.

- (a) What is a pure virtual function and an abstract class? Write a program demonstrating the use of virtual destructors. [7]
- (b) What is exception handling? Explain the use of `try`, `catch`, and `throw` blocks in C++ with a sample program. [7]

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